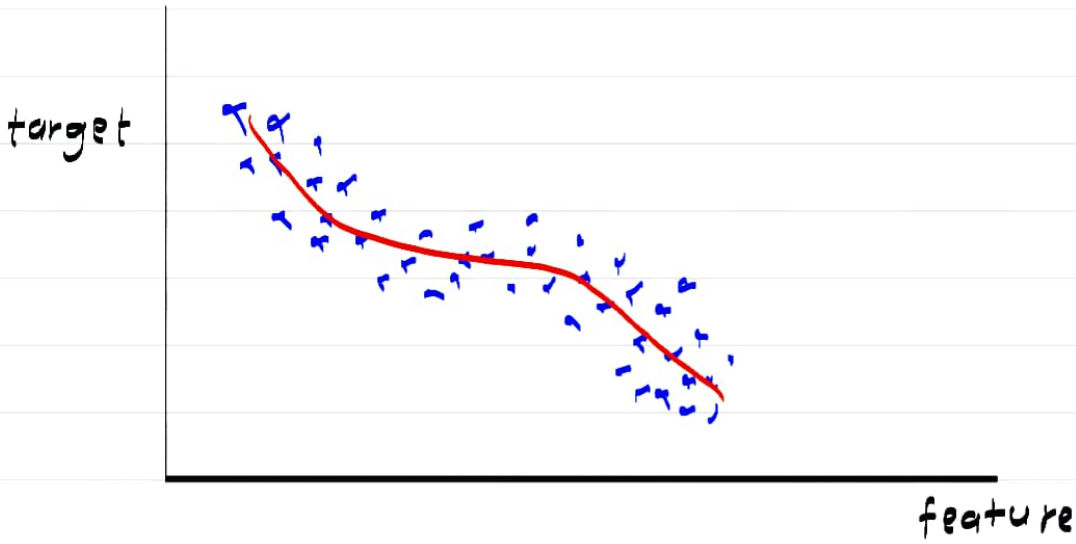
Machine Learning

13-04-26

Polynomial Regression



- linear regr won't work in this case
- polynomial regr. might work as relⁿ is non-linear

original data

x_1	y
1	1
2	2
4	5

$\xrightarrow{\text{polynomial degree} = 2}$

x_1	$x_1^2 = x_2$	y
1	1	1
2	4	2
4	16	5

$\downarrow$

$y = wx_1 + b$

$\downarrow$

$y = w_1 x_1 + w_2 x_2 + b$

$\downarrow$

find optimal values of w_1 and w_2 using gradient descent.

original data

x_1	x_2	y
1	1	10
2	4	12
3	6	14

$\xrightarrow[\text{degree} = 3]{\text{polynomial}}$

x_1	x_2	x_1^2	x_2^2	x_1^3	x_2^3	$x_1 x_2$	y
1	1	1	1	1	1	1	10
2	4	4	16	8	64	8	12
3	6	9	36	27	216	18	14

- polynomial degree is hyperparameter
- in most of the cases polynomial degree 2, 3 are used
  degree = 4, 5 are very uncommon.
- model training steps are similar to linear regression

Logistic Regression

- its used for binary classification
- logistic → as it uses log. function
- regression → as its output is continuous data
- supervised ML, parametric model

input → [Log regr] → output $\in [0, 1]$

$y = 0.5$
$y = 0.123$
$y = 0.789$
$y = 1$
$y = 0.9999$

apply threshold
thr: 0.7
if $y < 0.7$
output: A
if $y \geq 0.7$
output: B

x_1	x_2	x_3	y
⋮	⋮	⋮	A - +ve
⋮	⋮	⋮	B - -ve
⋮	⋮	⋮	A - +ve
			A - +ve
			B - -ve

only 2 classes in target hence this is binary clf data

eg.

weight	BP	blood sugar	diabetes
			yes
			no
			yes
			yes
			no

features

target

$x_1\ x_2\ x_3 \longrightarrow$ $\boxed{\text{Log regr}}$ $\longrightarrow$ logits $\xrightarrow{\text{apply threshold}}$ target

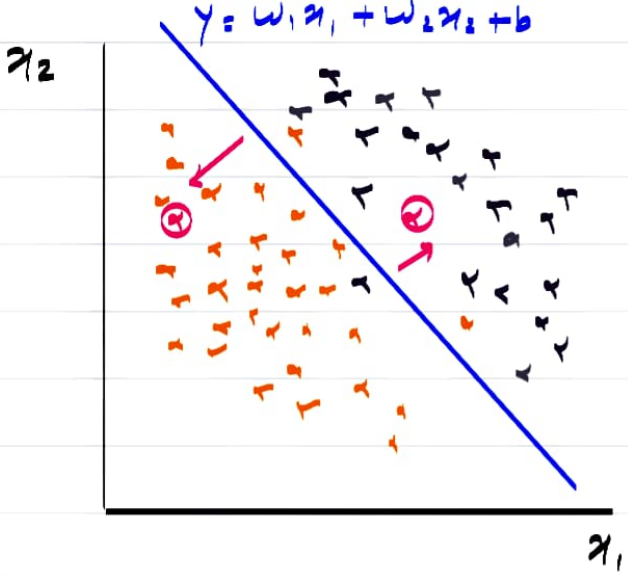
logits ↓ a real value betn 0 and 1

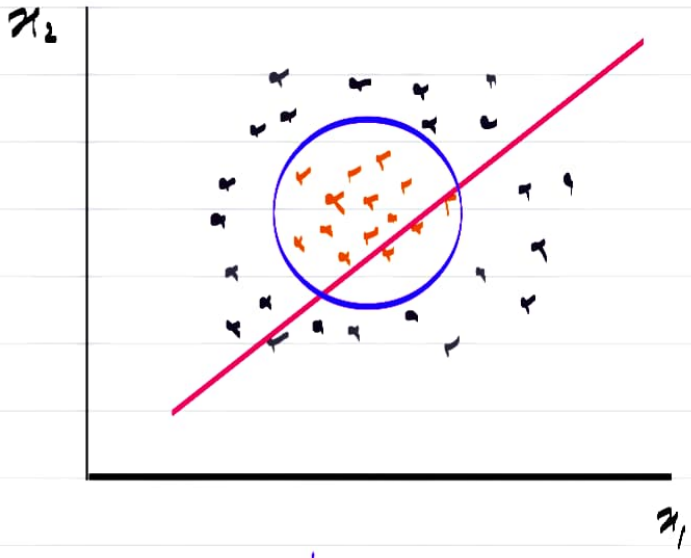
target: A or B, +ve or -ve

if logits $\geq$ threshold then $\hat{y}$ = +ve / A

if logits < threshold then $\hat{y}$ = -ve / B

- threshold value depends on the usecase in hand.
- we need to trail and test multiple values before finalizing the threshold.
- generally threshold values are greater than <u>0.5</u>

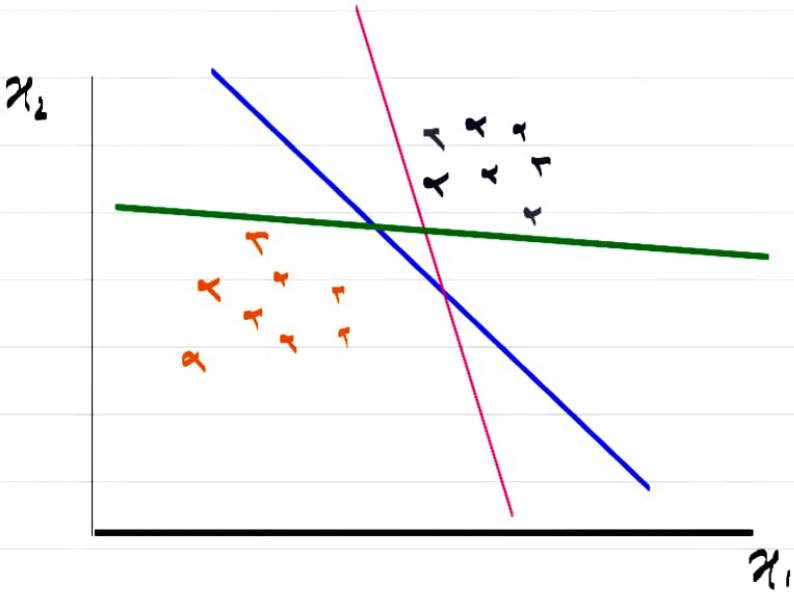




↓ log regr will work here

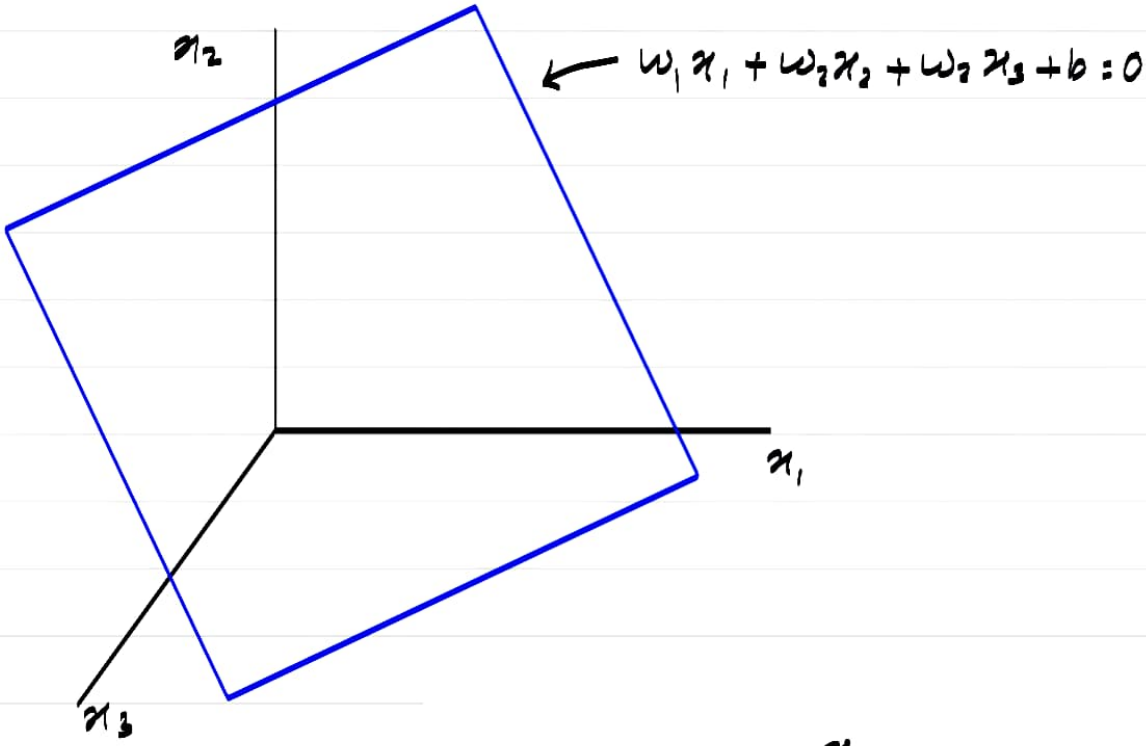
↓ log regr won't work here

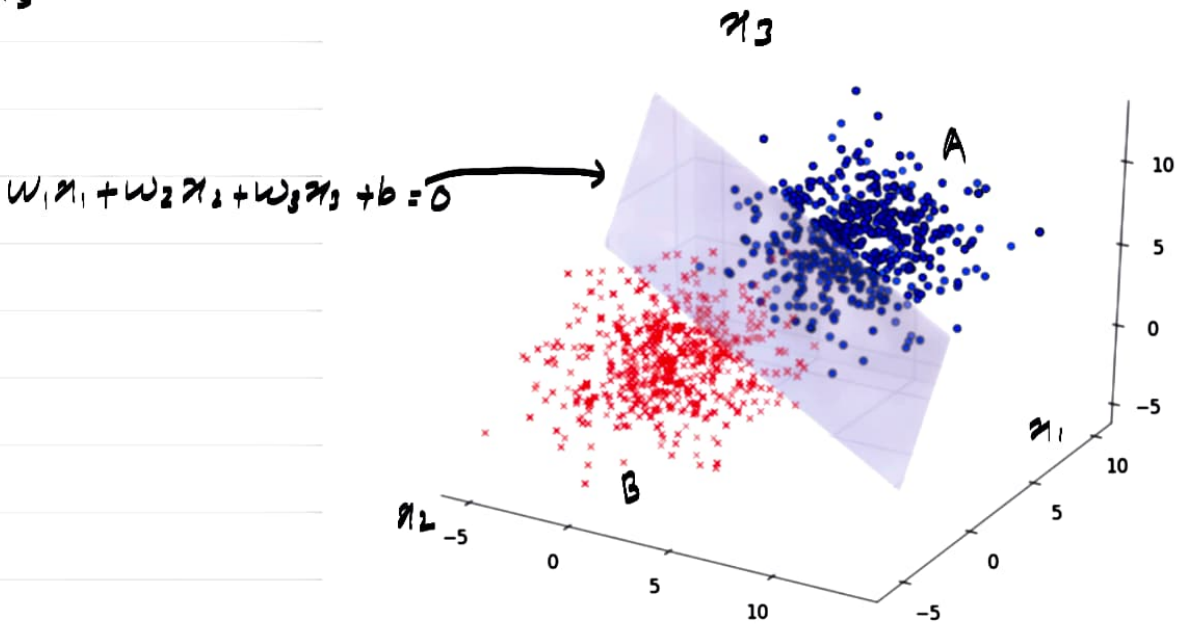
- Logistic regr. assumes that two classes are separable using a straight/linear decision boundary.



How to find the best line/plane which seperate two classes accurately?

x_2
x_1
$y = mx + c$
$mx + c - y = 0$
$w_1 x_1 + w_2 x_2 + b = 0$
general eqn of Line


x_2
x_1
x_3
$w_1 x_1 + w_2 x_2 + w_3 x_3 + b = 0$


x_3
$w_1 x_1 + w_2 x_2 + w_3 x_3 + b = 0$
A
B
x_1
x_2
10
5
0
−5
10
5
0
−5
−5
0
5
10

$x_1, x_2 \longrightarrow \boxed{\text{Log regr}} \longrightarrow y \in [0,1]$

$\downarrow$

+ve / -ve

y = probability = $P(o = +ve \mid x_1, x_2)$

$x_1 = 5$
$x_2 = 7$ $\longrightarrow \boxed{\text{Log regr}} \longrightarrow y = 0.78$

$0.78 = P(o = +ve \mid x_1 = 5 \text{ and } x_2 = 7)$

- conditional probability

- In log. regr. we want model to give prediction as probability $P(o = +ve \mid X)$

$\downarrow$

the prob. of this example being from +ve class given the input features X.

- In log. regr training we want our model to learn relationship betn features X and probability $P(o = +ve \mid X)$

eg.

$P(o = +ve \mid X) = 0.8$ ✓

then $P(o = -ve \mid X) = 1 - 0.8 = 0.2$

threshold: 0.65

$P(o = +ve \mid X) \geq$ threshold

$y = +ve$

How to find $p(y=+ve \mid X)$ from X

$$p(y=1 \mid X)$$

$$0 \leq p(y=1 \mid X) \leq 1$$

$$0 \leq p \leq 1 \longrightarrow \text{eqn } ①$$

eqn of linear decision boundary which seperates two classes

$$-\infty < w_1 x_1 + w_2 x_2 + w_3 x_3 + b < \infty \longrightarrow \text{eqn } ②$$

we need to connect these eqn ① and eqn ②

$$0 \leq p \leq 1$$

$$\boxed{0 \leq \frac{p}{1-p} \leq \infty}$$

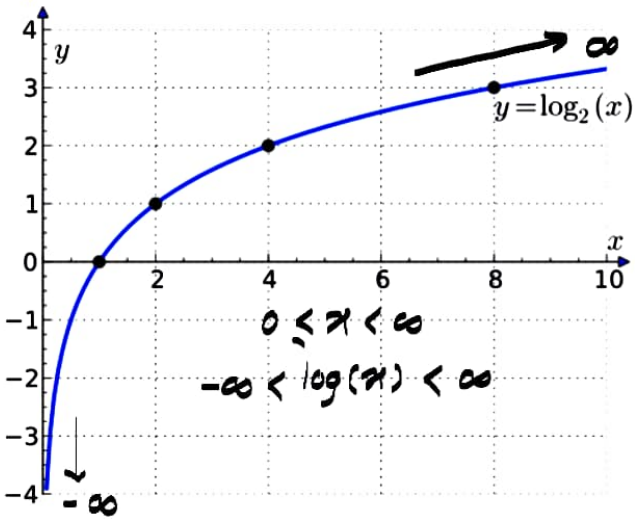
$\frac{p}{1-p} \rightarrow$ odds of an event

eg. $p = 0.5$ $\quad \frac{p}{1-p} = 1$

$p = 0.9$ $\quad \frac{p}{1-p} = \frac{0.9}{0.1} = 9$

$p = 0.2$ $\quad \frac{p}{1-p} = \frac{0.2}{0.8} = 0.25$

Log function



$$0 < \frac{p}{1-p} < \infty$$

$$-\infty < \log\left(\frac{p}{1-p}\right) < \infty$$

$$\updownarrow$$

$$-\infty < w_1 x_1 + w_2 x_2 + w_3 x_3 + b < \infty$$

$$\boxed{\log\left(\frac{p}{1-p}\right) = w_1 x_1 + w_2 x_2 + w_3 x_3 + b}$$